End Semester Examination, April – May, 2025

Degree: **BCA** Semester: **Fourth**

Stream: **BCA**

Paper Code: **BCACC401**

Paper Name: **Database Management Systems**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. Define the term "View of Data" in DBMS. **2**

2. What do you understand by physical data independence? **2**

3. Differentiate between schema and instance. **2**

4. What is SQL and why is it used? **2**

5. What is the difference between CHAR and VARCHAR data types? **2**

6. Explain the purpose of the GROUP BY clause. **2**

7. What is the difference between an entity and an entity set? **2**

8. Explain the concept of a weak entity. **2**

9. How does a dense index differ from a sparse index? **2**

10. What is a B-Tree index? **2**

11. What is a transaction? 2

12. What is the need for the COMMIT command? 2

13. What is a serial Transaction? 2

14. Why is a precedence graph needed? 2

15. What is a non-clustered index? 2

Part - B

Attempt any 6 questions out of 9 questions

Each question carries 5 Marks (5 × 6)

16. Compare and contrast file systems and databases in terms of security, redundancy, and integrity. 5

17. Explain the different types of Keys in the Relational Model with examples. 5

18. Differentiate between strong and weak entities with suitable examples. 5

19. Explain the difference between primary, secondary, and clustering indexes. 5

20. Explain the three levels of database architecture and how they relate to the view of data. 5

21. Explain the concept of Integrity Constraints in the Relational Model. 5

22. Describe the working of a dense index with an example. 5

23. Compare and contrast B-Tree and B+ Tree indexing. 5

24. Explain serial and non-serial schedules with proper examples. 5

Part - C

Attempt any 5 questions out of 8 questions

Each question carries 10 Marks (10 × 5)

25. Discuss in detail the advantages and disadvantages of using a database over a traditional file system. **10**

26. Write a relational algebra expression to find employees working in the 'HR' department and earning more than $50,000. Explain the steps. **10**

27. Explain in detail the steps involved in designing an ER Model for a university management system. **10**

28. Describe the process of creating and maintaining indexes in large databases. **10**

29. Compare hashing and tree-based indexing. Which one is better for large databases and why? **10**

30. Write the SQL statements using the university schema to perform the following operations: **10**

a) Create a new course "CS-001" titled "Weekly Seminar", with 0 credits.
b) Create a section of this course in Fall 2017, with sec_id of 1, and with the location of this section not yet specified.
c) Enroll every student in the comp. Sci. department in the above section.
d) Delete enrolments in the above section where the student's ID is 12345.
e) Delete the course CS-001. What will happen if you run this delete statement without first deleting offering (sections) of this course?
f) Delete all takes tuples corresponding to any section of any course with the word "advanced" as a part of the title; ignore case when matching the word with the title.

31. Explain Tuple Relational Calculus (TRC) with examples. **10**

32. Discuss the role of indexing in distributed databases and data warehouses. **10**

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